

Sentence Similarity II

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1. Problem Understanding

You are given:

- sentence1
- sentence2

Example

```
sentence1: I love aadya  
sentence2: I admire life
```

Also given similarPairs:

```
love <-> admire  
aadya <-> life
```

Need to determine: Can sentence1 be transformed into sentence2?

2. Key Concept: Transitivity

Similarity is transitive.

If:

$a \sim b$
 $b \sim c$
then
 $a \sim c$

Example Transformation:

aadya
↓
aadzz
↓
ethereal

Then: **aadya** $\sim$ **ethereal**

3. Approach: Disjoint Set Union (DSU)

DSU works on:

0... 1... 2...
Not strings.

Need mapping:

word → id

Therefore Use:

```
unordered_map<string, int> mp;
```

4. Implementation Code

```
class dsu {
public:
    vector<int> parent;
    vector<int> size;

    dsu(int n) {
        parent.resize(n);
        size.resize(n, 1);

        for(int i = 0; i < n; i++) {
            parent[i] = i;
        }
    }

    int findpar(int node) {
        if(parent[node] == node) {
            return node;
        }
        return parent[node] = findpar(parent[node]);
    }
}
```

```

void unionit(int u, int v) {
    int pu = findpar(u);
    int pv = findpar(v);

    if(pu == pv) {
        return;
    }

    if(size[pu] < size[pv]) {
        swap(pu, pv);
    }

    parent[pv] = pu;
    size[pu] += size[pv];
}

};

class Solution {
public:
    bool areSentencesSimilarTwo(vector<string>& sentence1,
                                vector<string>& sentence2,
                                vector<vector<string>>& similarPairs) {

        if(sentence1.size() != sentence2.size()) {
            return false;
        }

        unordered_map<string, int> mp;
        int id = 0;

        // har unique word ko unique id denge
        for(auto &it : similarPairs) {
            if(mp.find(it[0]) == mp.end()) {
                mp[it[0]] = id++;
            }
            if(mp.find(it[1]) == mp.end()) {
                mp[it[1]] = id++;
            }
        }

        dsu ds(id);
        for(auto &it : similarPairs) {
            int u = mp[it[0]];
            int v = mp[it[1]];
            ds.unionit(u, v);
        }

        for(int i = 0; i < sentence1.size(); i++) {
            if(sentence1[i] == sentence2[i]) {
                continue;
            }
        }
    }
};

```

```

    }

    // agar koi word tha hi nahi similiarity list me
    if(mp.find(sentence1[i]) == mp.end() ||
        mp.find(sentence2[i]) == mp.end()) {
        return false;
    }

    int u = mp[sentence1[i]];
    int v = mp[sentence2[i]];

    if(ds.findpar(u) != ds.findpar(v)) {
        return false;
    }
}

return true;
}

};

```